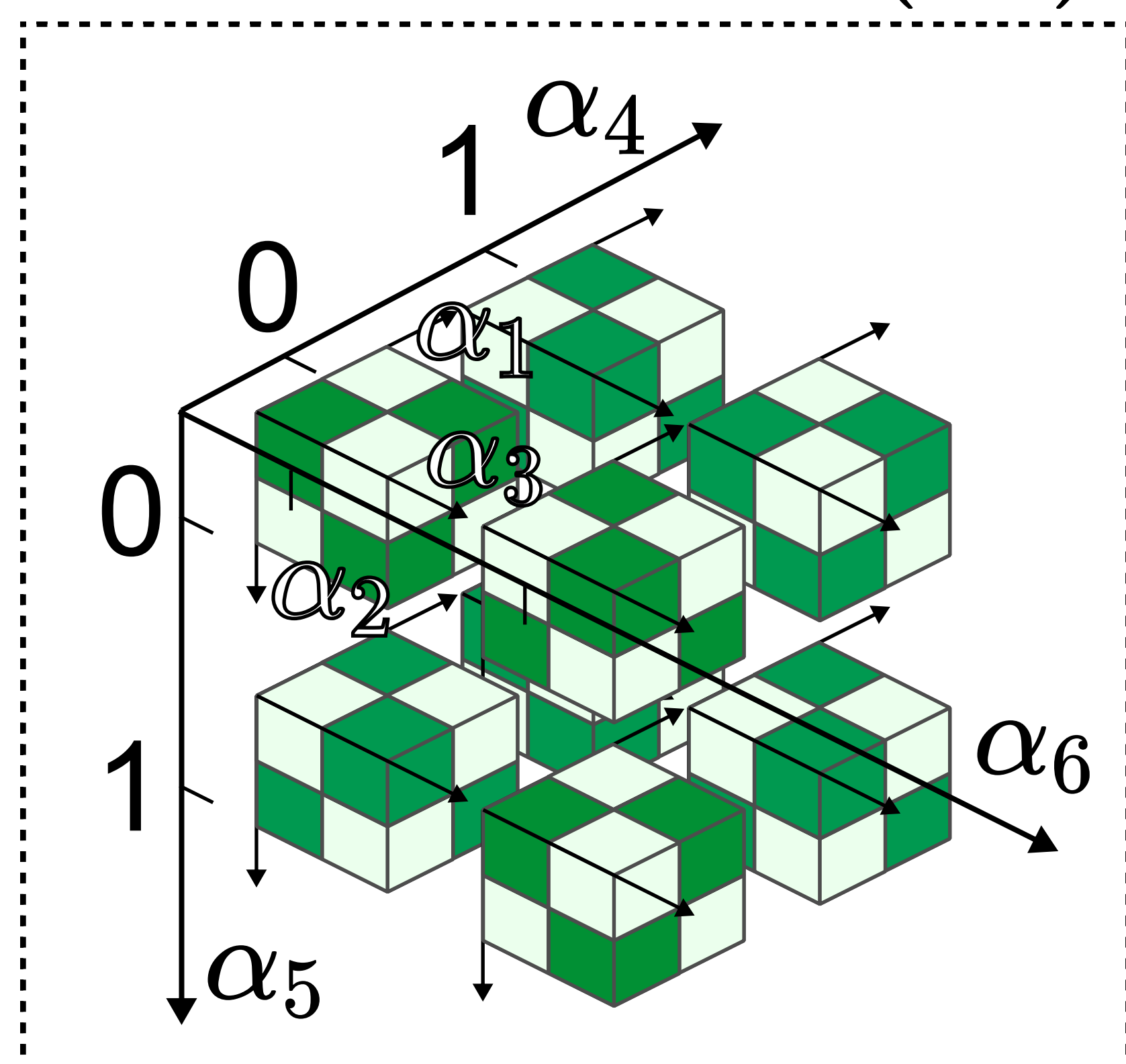


$$T_j[\alpha_1][\alpha_2] \cdots [\alpha_D] = \begin{array}{c} T_j^1 \\ \bullet \\ \alpha_1 \end{array} \text{---} \begin{array}{c} T_j^2 \\ \bullet \\ \alpha_2 \end{array} \cdots \begin{array}{c} T_j^{D-1} \\ \bullet \\ \alpha_{D-1} \end{array} \text{---} \begin{array}{c} T_j^D \\ \bullet \\ \alpha_D \end{array}$$

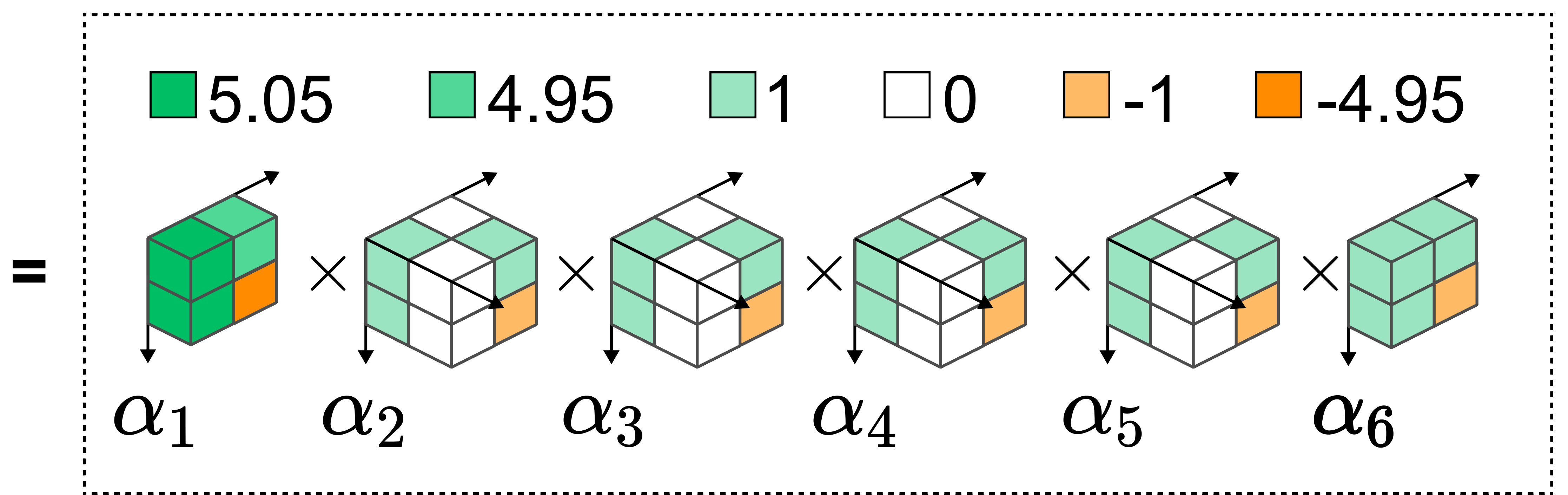
$$\cosh(w_j) + \sinh(w_j) \prod_k (-1)^{\alpha_k} \quad T_j^1[\alpha_1] = \begin{bmatrix} \cosh(\omega_j) \\ \sinh(\omega_j) \cdot (-1)^{\alpha_1} \end{bmatrix} \quad T_j^k[\alpha_k] = \begin{bmatrix} 1 & 0 \\ 0 & (-1)^{\alpha_k} \end{bmatrix} \quad T_j^D[\alpha_D] = \begin{bmatrix} 1 \\ (-1)^{\alpha_D} \end{bmatrix}$$

(a) Exact matrix product state decomposition of energy tensor

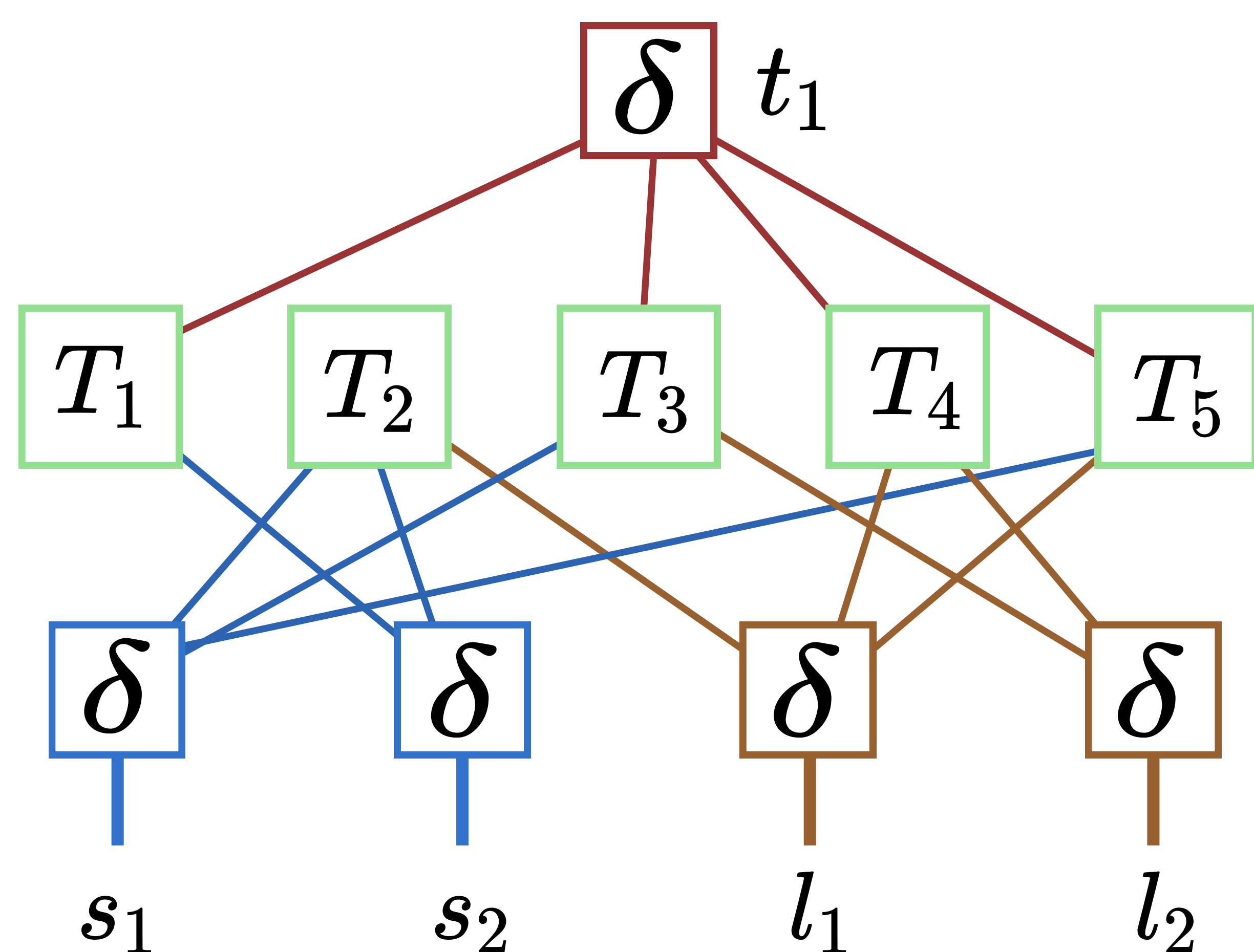
64 elements  $O(2^n)$



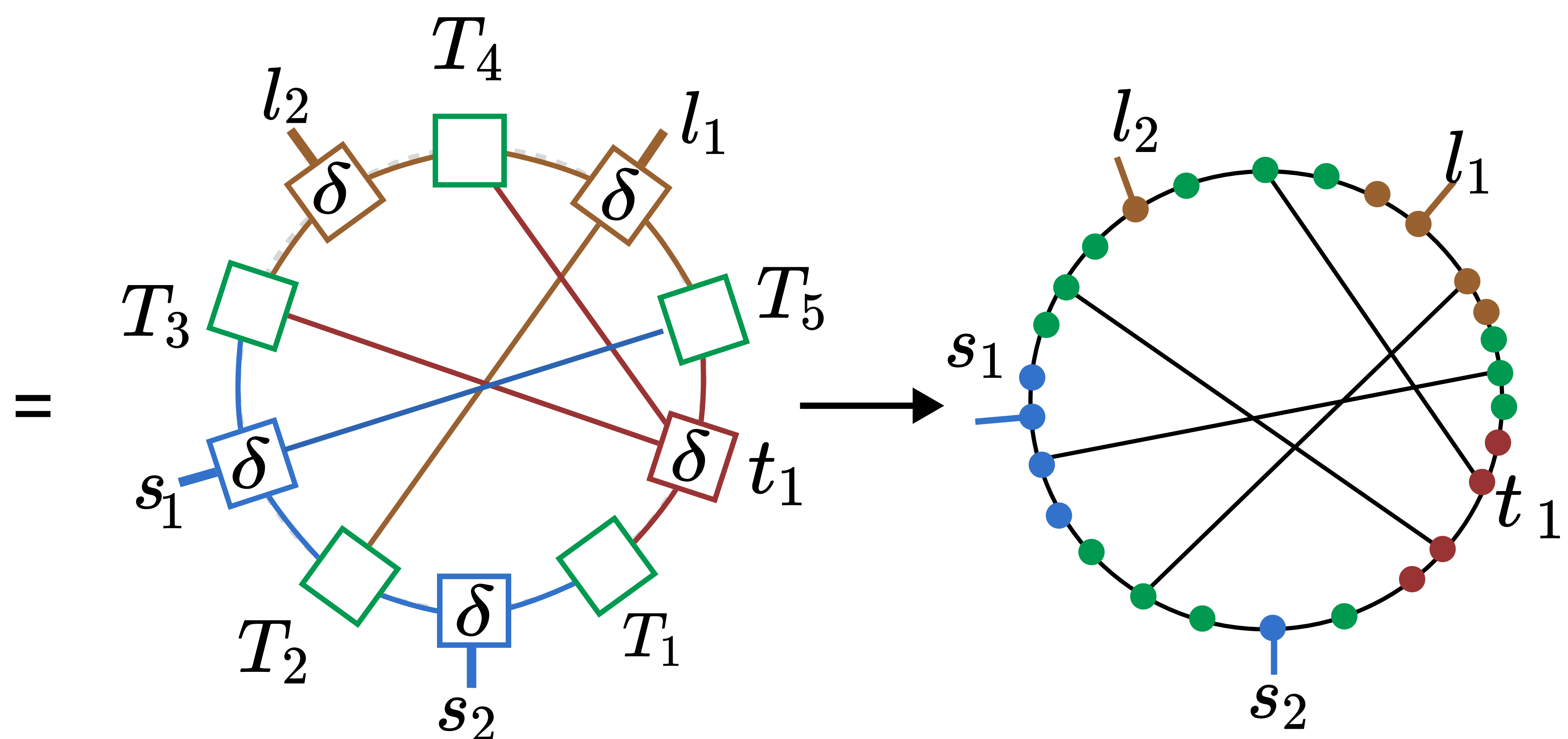
40 elements  $O(8n)$



(b) Example of energy tensor decomposition



(c) Example tensor network



(d) Circle layout

(e) Matrix product state